Model reduction for complex networks (and quantum systems)

– PRIN Kick off meeting –

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DIPARTIMENTO
DI INGEGNERIA
DELL'INFORMAZIONE

UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Agenda:

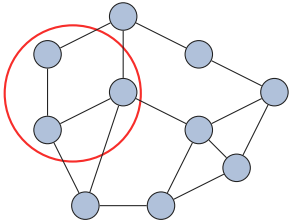
- Time-dependent reduced models;
- Perturbation theory – aka adiabatic elimination techniques;
- Symmetries;
- Model reduction for hidden Markov models (if we have time).
- Quantum stuff:
  - Aggregation methods;
  - Stochastic complement;

Some work in collaboration with M. Casanova, M. Peruzzo, F. Ticozzi (QuIDC group).

Time-dependent reduced models

arXiv:2510.20675

Time-dependent reduced models

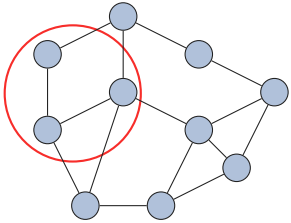


Consider a large, autonomous and **time-invariant** system.
Assume we are interested only in a small set of d.o.f., e.g. only a few nodes.

Question: Can we find an autonomous, **time-dependent** linear model that depends only on these few nodes that approximates well (or exactly reproduces) their dynamics?

Motivation: Simulation e.g. with Runge-Kutta methods.

Time-dependent reduced models



Let $P \in \mathbb{R}^{n\times n}$ be the projector onto these d.o.f.
i.e. $P^2 = P$, $\mathscr{V} \equiv \mathrm{Im}P$ with $m = \dim(\mathscr{V})$ and $m << n$.
Let $P = JR$ with $R \in \mathbb{R}^{m\times n}$ and $J \in \mathbb{R}^{n\times m}$, e.g. $R = [\mathbb{1}_m | 0_{n-m}]$.

Assume: $Px_0 = x_0$.

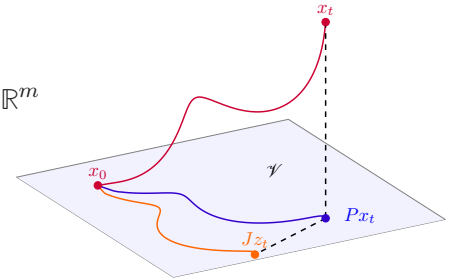
$$\Sigma_{TI} : \begin{cases} \dot{x}_t = Lx_t \\ y_t = Rx_t \end{cases} \qquad x \in \mathbb{R}^n,\ y \in \mathbb{R}^m.$$

Task: Find

$$\Sigma_{TD} : \begin{cases} \dot{z}_t = F_t z_t \\ z_0 = Rx_0 \end{cases} \qquad z \in \mathbb{R}^m$$

such that

$$y_t = z_t, \qquad \forall t \geq 0.$$



Nakajima-Zwanzig equations (1958-1960)

Let $Q \equiv \mathbb{1} - P$, i.e. the projector on "the rest". Then:

$$\frac{d}{dt} P x_t = PLPx_t + PLQx_t, \tag{1}$$

$$\frac{d}{dt} Q x_t = QLPx_t + QLQx_t. \tag{2}$$

Using Lagrange's equation to solve (2) we have

$$Qx_t = e^{QLQt} Qx_0 + \int_0^t e^{QLQs} QLPx_{t-s} ds$$

which, when inserted into (1), gives

$$\frac{d}{dt} P x_t = PLPx_t + PLe^{QLQt} Qx_0 + \int_0^t PLQe^{QLQs} QLPx_{t-s} ds. \tag{3}$$

Time convolutionless Nakajima-Zwanzig equations

Recall that $x_{t-s} = e^{-Ls}x_t$ so that

$$Qx_t = e^{QLQt}Qx_0 + \int_0^t e^{QLQs}QLPe^{-Ls}ds \cdot (P+Q)x_t.$$

Now, defining

$$M_t \equiv \mathbb{1} - \int_0^t e^{QLQs}QLPe^{-Ls}ds \tag{4}$$

we have

$$M_tQx_t = e^{QLQt}Qx_0 + (\mathbb{1} - M_t)Px_t.$$

If we then assume that M_t is invertible (typically holds for small t), we obtain

$$Qx_t = M_t^{-1}e^{QLQt}Qx_0 + (M_t^{-1} - \mathbb{1})Px_t.$$

Time convolutionless Nakajima-Zwanzig equations

Substituting this into Eq. (1) we obtain

$$\frac{d}{dt}Px_t = PLM_t^{-1}e^{QLQt}Qx_0 + PLM_t^{-1}Px_t. \tag{5}$$

Note that under the assumption $Px_0 = x_0$ we have $Qx_0 = 0$ and, by separating the projector P into its factors $P = JR$, we obtain

$$\frac{d}{dt}\check{x}_t = \underbrace{RLM_t^{-1}J}_{\equiv F_t}\check{x}_t. \tag{6}$$

When M_t is invertible, this provides an **exact solution.**
Problem: Impractical to compute (at least analytically) since we need to compute an integral and an inverse.

A perturbative approach

Fact: F_t is analytic in t, i.e. $F_t = \sum_{k=0}^{+\infty} t^k F_{(k+1)}$.

Can we find an approximation with a truncation, i.e. $F_{t,N} = \sum_{k=0}^{N} t^k F_{(k+1)}$?
How could we compute the terms $F_{(k+1)}$?

Theorem: Let $F_t = \sum_{k=0}^{+\infty} t^k F_{(k+1)}$ with $F_{(k)} = 0$ for all $k > N$. Then the flow

$$\mathbb{T} e^{\int_0^t F_{s,N} ds} = \sum_{k=0}^{+\infty} t^k E_{(k)} \qquad \text{with} \qquad E_{(k)} = \frac{1}{k} \sum_{s=1}^{k} F_{(s)} E_{(k-s)}$$

and $E_{(0)} = \mathbb{1}$.

A perturbative approach

The trick is simple:

$$Re^{Lt}J = \sum_{k=0}^{+\infty} \frac{t^k}{k!} RL^k J \qquad \approx \qquad \sum_{k=0}^{+\infty} t^k E_{(k)} = \mathbb{T}e^{\int_0^t F_{s,N} ds}$$

we want to match the terms $E_{(k)}$ and $\frac{RL^k J}{k!}$ up to a certain order N.

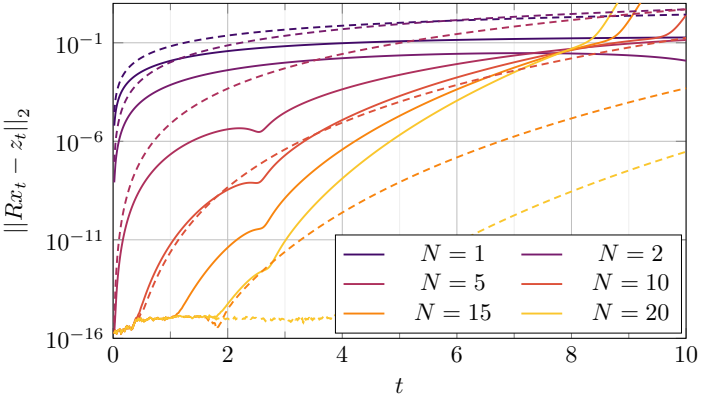
Theorem: Let

$$F_{t,N} \equiv \sum_{k=0}^{N} t^k F_{(k+1)} \qquad \text{where} \qquad F_{(k)} \equiv k\frac{RL^k J}{k!} - \sum_{h=1}^{k-1} F_{(k-h)} \frac{RL^h J}{h!}.$$

Then:

$$Re^{Lt}J - \mathbb{T}e^{\int_0^t F_{s,N} ds} = O(t^{N+1}).$$

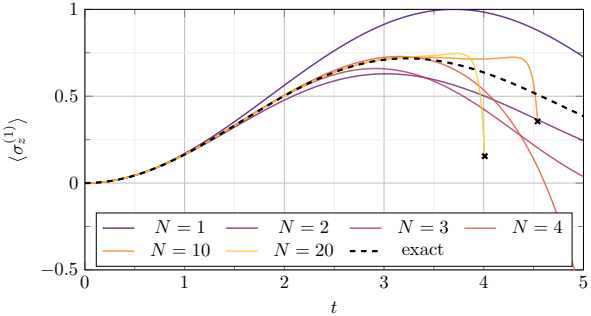
Results



At low orders we perform better than a truncation of the exponential (dashed line).

Outlook:

- In simulations we observe Runge's phenomena:



  Can Padé's approximations help?

- Can we reverse the process to prove properties of time-varying systems? That is: *Given a LTV system find an equivalent, large, LTI system.*

Perturbation theory (aka adiabatic elimination)

Work in progress (to appear soon).
Joint work with Alain Sarlette, Francesco Ticozzi and Lorenza Viola.

The problem

Consider a linear dynamical model that is **easy to study**:

$$\dot{x}_t = L_{(0)} x_t$$

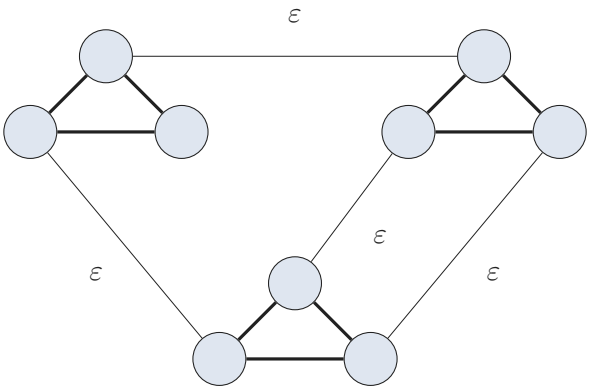
and assume that such a model is analytically perturbed with a small parameter ε:

$$\dot{x}_t = L_\varepsilon x_t, \qquad L_\varepsilon = \sum_{k=0}^{+\infty} \varepsilon^k L_{(k)}.$$

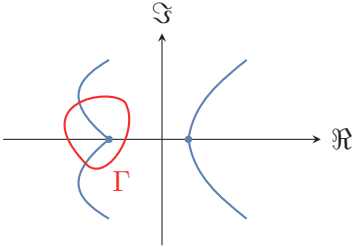
We want to approximate the dynamics of the perturbed generator L_ε around “interesting” eigenvalues of $L_{(0)}$, e.g. the equilibria.

Example

Consider a network made of identical sub-networks connected by random weak links.



Perturbation theory – Kato (1995)



The eigenvalues of L_ε are branches of analytic functions in ε. If we group a set of them together we can find, for ε small enough, an analytic spectral projection P_ε, i.e.

$$P_\varepsilon^2 = P_\varepsilon, \quad P_\varepsilon L_\varepsilon = L_\varepsilon P_\varepsilon, \quad P_\varepsilon = \sum_{k=0}^{+\infty} \varepsilon^k P_{(k)}$$

with $P_{(0)}$ the spectral projector on the corresponding eigenvalues of $L_{(0)}$. The projector is formally defined as

$$P_\varepsilon = -\frac{1}{2\pi i} \int_\Gamma (L_\varepsilon - z)^{-1} dz$$

where Γ is a curve enclosing all the eigenvalues of interest.

Good luck in computing this in practice.

Adiabatic elimination – the idea

Once again, compute the terms P_k of P_ε iteratively by imposing

$$\begin{cases} P_\varepsilon^2 = P_\varepsilon \\ [P_\varepsilon, L_\varepsilon] = 0 \end{cases} \quad \Leftrightarrow \quad \forall k \begin{cases} \sum_{j=0}^{k} P_j P_{k-j} = P_k \\ \sum_{j=0}^{k} [P_j, L_{k-j}] = 0 \end{cases}$$

Theorem: Assume

$$L_0 = \left[\begin{array}{c|c} L_0^S & 0 \\ \hline 0 & L_0^R \end{array}\right], \quad P_0 = \left[\begin{array}{c|c} \mathbb{1} & 0 \\ \hline 0 & 0 \end{array}\right], \quad L_j = \left[\begin{array}{c|c} L_j^S & L_j^P \\ \hline L_j^Q & L_j^R \end{array}\right], \qquad P_j = \left[\begin{array}{c|c} P_j^S & P_j^P \\ \hline P_j^Q & P_j^R \end{array}\right].$$

Then, the terms of the projector P_k are completely determined by

$$\begin{cases} P_k^P L_0^R - L_0^S P_k^P = -L_k^P + \sum_{j=1}^{k-1} \left(L_{k-j}^P P_j^R + L_{k-j}^S P_j^P - P_j^P L_{k-j}^R - P_j^S L_{k-j}^P \right) \\ P_k^Q L_0^S - L_0^R P_k^Q = L_k^Q + \sum_{j=1}^{k-1} \left(L_{k-j}^Q P_j^S + L_{k-j}^R P_j^Q - P_j^Q L_{k-j}^S - P_j^R L_{k-j}^Q \right) \end{cases} \tag{7}$$

$$\begin{cases} P_k^S = -\sum_{j=1}^{k-1} (P_j^S P_{k-j}^S + P_j^P P_{k-j}^Q) \\ P_k^R = \sum_{j=1}^{k-1} (P_j^R P_{k-j}^R + P_j^Q P_{k-j}^P) \end{cases} . \tag{8}$$

Adiabatic elimination – Factorization

By computing a factorization of the projector $P_\varepsilon = J_\varepsilon R_\varepsilon$ we can compute the reduced model onto $\mathrm{Im} P_\varepsilon$.

$$\check{L}_\varepsilon = R_\varepsilon L_\varepsilon J_\varepsilon$$

such that

$$e^{L_\varepsilon t} x_0 = J_\varepsilon e^{\check{L}_\varepsilon t} R_\varepsilon x_0, \qquad \forall t \geq 0,\ \forall x_0 \in \mathrm{Im} P_\varepsilon.$$

Conjecture: Assume

$$R_0 = \left[\begin{array}{c|c} \mathbb{1} & 0 \end{array}\right], \qquad J_0 = \left[\begin{array}{c} \mathbb{1} \\ \hline 0 \end{array}\right], \qquad R_k = \left[\begin{array}{c|c} R_k^A & R_k^B \end{array}\right], \qquad J_k = \left[\begin{array}{c} J_k^A \\ \hline J_k^B \end{array}\right].$$

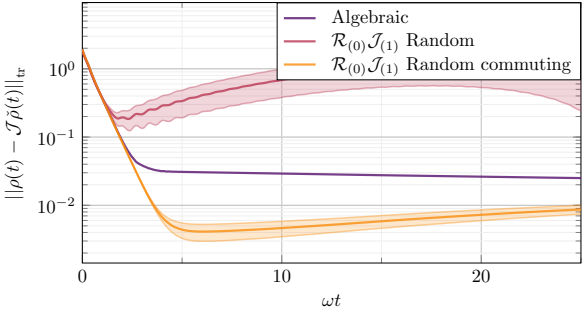
Then J_k^A is a **free variable**.
[**Note**: Proven for orders 1 and 2. Working on the higher orders.]

Adiabatic elimination – Gauge degrees of freedom

By truncating the reduced model at order 1 we obtain

$$\tilde{L}_1 = L_0^S + \varepsilon \left(L_1^S + [L_0^S, J_1^A] \right) .$$

Open question: How to choose J_1^A so that $\tilde{L}_1$ approximates best the evolution of $\check{L}_\varepsilon$?



Symmetries

Symmetries

Consider a dynamical semigroup $\{e^{Lt}\}$.
Symmetries are useful to determine invariant subspaces of the dynamics:

Def: an invertible U is a symmetry for the generator L, if $[L, U] = 0$.
Property: **Any generalized eigenspace of U is L-invariant**.
Question: *In networks,* can we identify symmetries from locality?

In quantum systems: the generator is a linear map $\mathcal{L} : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}$ that satisfy certain properties. A symmetry in this context takes the form $\mathrm{Ad}_U(X) = UXU^\dagger$, with U unitary.

Question: How to find symmetries for a given generator $\mathcal{L}$?
Solution: **Riemannian optimization**

$$\min_{U\in SU(n)} ||[\mathcal{L}, \mathrm{Ad}_U]||^2$$

(Work in progress with M. Casanova, L. Viola and F. Ticozzi)

Model reduction for hidden Markov models

Joint work with Francesco Ticozzi.
arXiv:2208.05968v2 (IEEE TAC)

Hidden Markov models (a version)

$\mathbb{R}^n$

p are probability distributions:
$p \in \mathbb{R}^n$, $p_i \geq 0$, $\sum_i p_i = 1$

P is a **stochastic matrix**

$\mathcal{S}$ is a set of initial conditions

$$\begin{cases} p(t+1) = \boxed{P}\,[\,\boxed{p(t)}\,] \\ \boxed{y(t)} = \boxed{C}\,[p(t)] \end{cases}, \qquad \boxed{p_0 \in \mathcal{S}}$$

$y(t) \in \mathbb{R}^m$ is the output of interest, the one we want to preserve

C is a linear output map (could be stochastic)

The problem: Positive-preserving model reduction

Given a model $(P, \mathcal{S}, C)$ find **another probability model** $(\check{P}, \check{\mathcal{S}}, \check{C})$ and a linear map $\Phi : \mathbb{C}^{n\times n} \to \mathbb{C}^{r\times r}$

such that for all $t \geq 0$ and $p_0 \in \mathcal{S}$, $\check{p}_0 = \Phi[p_0]$

- *Exact model reduction* (almost done)

$$CP^t p_0 = \check{C}\check{P}^t \check{p}_0;$$

- *Approximate model reduction* (ongoing work)

$$CP^t p_0 \approx \check{C}\check{P}^t \check{p}_0.$$

Note: another constraint we might consider is the fact $\check{P}$ preserves the locality of the graph generated by P (we can only join together nodes, not create new edges).

Exact MR - a bignami

MR based on the Non-Observable space

Assume

$$\ker[C] \neq \{0\},$$

$$p_0 \text{ any prob. vect.}$$

We are only interested in reproducing trajectories of certain values of interest starting from any initial condition.

Note: One can study the dual problem as well.

Linear model reduction

$$\begin{cases} p(t+1) = P[p(t)] \\ y(t) = C[p(t)] \end{cases}$$

$$\mathscr{N} = \{x \in \mathbb{R}^n | \quad CP^t x = 0, \forall t \geq 0\}$$

$\mathscr{N}$ is the largest P-invariant operators subspace contained in $\ker[C]$.

$$\begin{cases} \left[\begin{array}{c} p_{\mathscr{N}} \\ \hline p_{\mathscr{N}^\perp} \end{array}\right](t+1) = \left[\begin{array}{c|c} P_{\mathscr{N},\mathscr{N}} & P_{\mathscr{N},\mathscr{N}^\perp} \\ \hline 0 & P_{\mathscr{N}^\perp,\mathscr{N}^\perp} \end{array}\right] \left[\begin{array}{c} p_{\mathscr{N}} \\ \hline p_{\mathscr{N}^\perp} \end{array}\right](t) \\ y(t) = \left[\begin{array}{c|c} 0 & C_{\mathscr{N}^\perp} \end{array}\right] \left[\begin{array}{c} p_{\mathscr{N}} \\ \hline p_{\mathscr{N}^\perp} \end{array}\right](t) \end{cases}$$

Linear model reduction pt. 2

This is equivalent to finding factors $R \in \mathbb{R}^{m \times n}$ and $J \in \mathbb{R}^{n \times m}$ such that $JR = \Pi_{\mathscr{N}^\perp}$ and $RJ = I_m$ (where $m = \dim(\mathscr{N}^\perp)$) so that

$$\check{P} = RPJ, \qquad \check{C} = CJ, \qquad \Phi = R.$$

Note for later:
Restricting onto $\mathscr{N}^\perp$ provides the minimal linear equivalent model.
Nonetheless, we can include some variables from $\mathscr{N}^\perp$ and the reduced model is still equivalent.

Problem: $\check{P}$ is not necessarily stochastic

Algebras and stochastic projectors

Consider now a vector space $\mathscr{A}$ that is also closed w.r.t. element wise product: $v, w \in \mathscr{A}$, $v \wedge w \in \mathscr{A}$.
Then $\mathscr{A} = \mathrm{span}\{a_j\}$ with idempotent vectors a_j, i.e. $a_j \wedge a_k = a_j \delta_{j,k}$.

Consider then a probability vector $q \in \mathbb{R}$ with full support and define

$$\Pi \equiv \sum_j \frac{(q \wedge a_j) a_j^T}{\langle q, a_j \rangle}.$$

Π is a *stochastic* projector onto $\mathrm{Im}(\Pi) = q \wedge \mathscr{A}$ and orthogonal w.r.t

$$\langle x, y \rangle_{q^{-1}} = x^T \mathrm{diag}(q)^{-1} y.$$

Probabilistic intuition

Algebras are associated to σ-algebras.
Given a set of r.v.s (vectors) $\{v_j\}$, closing it to an algebra is equivalent to finding the smallest σ-algbera that makes each r.v. measurable.

Given an algebra $\mathscr{A}$ and a probability distribution q, we can call $\mathbb{E}_{|\mathscr{A},q}$ the conditional expectation w.r.t. q and conditioned to the σ-algebra associated to $\mathscr{A}$. Then

$$\Pi = \mathbb{E}^T_{|\mathscr{A},q}$$

is the dual of a cond. exp. (the projector for states instead of the projector for random variables).

Factorization of Π

$(q \wedge \mathscr{A}) \simeq \mathbb{R}^h$, let $\{e_j\}$ be the standard basis for $\mathbb{R}^h$. Define:

$$J = \sum_j \frac{(q \wedge a_j) e_j^T}{\langle q, a_j \rangle} \in \mathbb{R}^{n \times h}, \quad R = \sum_j e_j a_j^T \in \mathbb{R}^{h \times n}$$

so that $\Pi = JR$ and $RJ = I_h$.

R and J are stochastic.

Note: R is a matrix made of 0s and 1s and is stochastic, i.e. its a clustering.

Observable positive reduction

We leverage the knowledge of the quantities of interest to reduce the model.

1. Find the **non-observable subspace** $\mathscr{N} = \{x \in \mathbb{R}^n | \quad CP^t x = 0, \forall t \geq 0\}$
2. Close its orthogonal complement to an **algebra** $\mathscr{A} = \mathrm{alg}(\mathscr{N}^\perp)$;
3. Choose q (how? uniform dist. is ok) and compute Π and its two factors R and J;
4. Use the two factors to **reduce the dynamics**:

$$\check{P} = RPJ, \qquad \check{C} = CJ, \qquad \Phi = R.$$

Note: $\check{P}$ is stochastic and retains the locality of the network.
Main result: The closure of $\mathscr{N}^\perp$ (w.r.t. $\langle \cdot, \cdot \rangle_{HS}$) to an algebra is "optimal".

Reachable positive reduction

One can similarly solve the problem for the dual, i.e. $C = I_n$ and a subset of p_0 leveraging the knowledge of initial conditions.

1. Find the **reachable subspace** $\mathscr{R} = \mathrm{span}\{P^t p_0 = 0, \forall t \geq 0\}$
2. Choose q (we know how) and compute the algebra $\mathscr{A} = \mathrm{alg}(q^{-1} \wedge \mathscr{R})$;
3. Compute Π and its two factors R and J;
4. Use the two factors to **reduce the dynamics**:

$$\check{P} = RPJ, \qquad \check{C} = CJ, \qquad \Phi = R.$$

Open problem: Optimality of the iterative procedure.

From Exact to Approximate MR

Issues:

- P **is not stable:** Moment matching methods can not be used;
- Closing numerically generated spaces (e.g. SVD) to algebras generates everything.

In the literature:

- Positive-preserving MR (no stochastic matrices);
- Locality preserving MR (often not positivity preserving);
- Markov chains MR (mostly cares about specific metrics).

Other possible connections:

- Kron-reduction (and stochastic complements, see below);
- Perturbation methods (quantum adiabatic elimination, see above).

Asymptotically Exact MR (done!)

Let us relax *exact MR* to *asymptotically exact MR* (exact for $t \to +\infty$):

- **Fact 1:** Any asymptotically exact reduction of a linear system must contain the center manifold;
- **Fact 2:** The "center manifold" of a Markov chain is an algebra;
- **Then:** We can project on that and obtain the *minimal* asymptotically exact model.

Issues: it can be hard to compute, especially for large systems.

Perturbative approach: if we can consider the model of interest as the perturbation of one with a simple center manifold (e.g. block diagonal), then we can project there.

Desiderata

Approx model reduction such that:

- Stochasticity preserving (or positive, at least);
- Locality preserving;
- Computable from local information.

Quantum stuff

Symmetries

Consider a dynamical semigroup $\{e^{Lt}\}$.
Symmetries are useful to determine invariant subspaces of the dynamics:

Def: an invertible U is a symmetry for the generator L, if $[L, U] = 0$.
Property: **Any generalized eigenspace of U is L-invariant**.
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Question: How to find symmetries for a given generator $\mathcal{L}$?
Solution: **Riemannian optimization**

$$\min_{U\in SU(n)} ||[\mathcal{L}, \mathrm{Ad}_U]||^2$$

(Work in progress with M. Casanova, L. Viola and F. Ticozzi)

Other open projects on quantum systems

- Symmetries for quantum trajectories (stochastic, monitored systems);
- We are extending aggregation (clustering) methods used for networks to quantum systems;
- We are extending classical **stochastic complement theory** to hybrid quantum-classical and full quantum models.
  It provides:
  - a way to decompose and find equilibria *one sub-network at the time*;
  - a Simon-Ando theory showing separation of time scales, if the subnetworks are weakly connected.

Thanks for your attention!

Stochastic Complement and Uncoupling of Markov Chains

A review.

Setup: partitioned Markov chain

Let P be the transition matrix of a finite Markov chain, partitioned into blocks:

$$P = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1k} \\ P_{21} & P_{22} & \cdots & P_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ P_{k1} & P_{k2} & \cdots & P_{kk} \end{bmatrix}, \qquad P\mathbf{1} = \mathbf{1}.$$

Interpretation:

- each block corresponds to a subset of states;
- P_{ii} describes transitions that remain within block i;
- off-diagonal blocks P_{ij} describe communication between blocks.

Goal: study this chain one block at the time, or block-wise.

Stochastic complement

For a fixed block i, write

$$P = \begin{bmatrix} P_{ii} & P_{i*} \\ P_{*i} & P_i \end{bmatrix}.$$

The **stochastic complement** of P_{ii} in P is

$$S_{ii} = P_{ii} + P_{i*}(\mathbb{1} - P_i)^{-1} P_{*i}.$$

Equilibrium condition: it can be shown directly that $p = [p_i, q_i]^T$ is an equilibrium if and only if $S_{ii} p_i = p_i$.

Probabilistic meaning: notice that $S_{ii} = P_{ii} + P_{i*}P_{*i} + P_{i*}P_iP_{*i} + P_{i*}P_i^2P_{*i} + \cdots$.
As if we start in block i; allow an excursion outside block i; return eventually to block i.

Fact: it is a stochastic matrix.

Coupling matrix and exact aggregation

Define the block-diagonal matrix of stochastic complements

$$S = \mathrm{diag}(S_{11}, \ldots, S_{kk}).$$

Then:

- S is block diagonal, so its classes are decoupled;
- if P is irreducible, each S_{ii} has a single equilibrium.
- within each block, S_{ii} preserves the reduced equilibria induced by P (up to a rescaling);

Aggregation viewpoint: the chain on all states can be decomposed into *within-block equilibration* plus *between-block coupling*;

Nearly uncoupled chains and Simon–Ando idea

Suppose

$$P = P^{(0)} + \varepsilon C,$$

where $P^{(0)}$ is block diagonal and ε is small.
Then:

- each diagonal block mixes rapidly on its own;
- communication between different blocks is weak;
- the spectrum splits into:
  - several eigenvalues near 1 (slow inter-block motion),
  - the remaining eigenvalues bounded away from 1 (fast intra-block decay).

This is the core Simon–Ando picture:

fast local relaxation $\Longrightarrow$ metastable plateaus $\Longrightarrow$ slow global equilibration.

Key Simon–Ando conclusions

For a nearly completely reducible Markov chain:

1. **Short run:** probability mass rapidly approaches quasi-equilibrium within each block. Correspond to stochastic complement equilibria;
2. **Middle run:** the chain behaves approximately like a reduced Markov chain on the blocks;
3. **Long run:** the full chain converges to its unique global stationary distribution.